A Project Report on

“Sales Data Analysis and visualization”

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PROJECT OVERVIEW

Introduction

Data analysis is an important process used to extract useful insights from raw data. Businesses use data analysis to understand customer behavior, sales patterns, and product performance. Visualization techniques help represent complex data in a simple and understandable way.

In this project, a sales dataset is analyzed using Python libraries such as Pandas and Matplotlib. The dataset contains information about products, quantity sold, price, region, and total sales.

The main aim of this project is to analyze sales performance and visualize patterns in the dataset using different charts.

Objectives

The objectives of this project are:

- To load and explore a sales dataset using Python.
- To perform basic data analysis on product sales.
- To visualize sales data using charts.
- To identify patterns trends in sales data.

SETUP INSTRUCTIONS

To run this project, the following steps should be followed.

Step 1: Install Required Libraries
The following Python libraries are required:

- Pandas
- Matplotlib

Installation command:
Pip install pandas matplotlib

Step 2: Import Libraries
The required libraries are imported in Python as follows:

- Import pandas as pd
- Import matplotlib.pyplot as plt

Step 3: Load Dataset

The dataset is loaded using the Pandas library.

- Data = pd.read_csv("sales_data.csv")

CODE STRUCTURE

```python
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
data = pd.read_csv("/content/sales_data (2).csv")

# Dataset preview
print(data.head())

# Check missing values
print(data.isnull().sum())

# Basic info
print(data.info())

# 1 Product wise total sales
product_sales = data.groupby("Product")["Total_Sales"].sum()

plt.figure()
product_sales.plot(kind="bar")
plt.title("Total Sales by Product")
plt.xlabel("Product")
plt.ylabel("Total Sales")
plt.show()
```

```python
# 2 Region wise sales
region_sales = data.groupby("Region")["Total_Sales"].sum()

plt.figure()
region_sales.plot(kind="pie", autopct='%1.1f%%')
plt.title("Sales Distribution by Region")
plt.ylabel("")
plt.show()

# 3 Quantity sold by product
quantity_sales = data.groupby("Product")["Quantity"].sum()

plt.figure()
quantity_sales.plot(kind="line")
plt.title("Quantity Sold by Product")
plt.xlabel("Product")
plt.ylabel("Quantity")
plt.show()

# Insights
print("Highest selling product:", product_sales.idxmax())
print("Lowest selling product:", product_sales.idxmin())
```

The project code is organized into several sections for clarity and better understanding.

- Data Loading
  The dataset is loaded into a Pandas DataFrame.
- Data Exploration
  Basic information about the dataset is checked using functions such as:
  Head()
  Info()
  Isnull()

This helps understand the structure and content of the dataset.

- Data Analysis
  Data analysis is performed using groupby operations to calculate total sales by product and region.

Example:

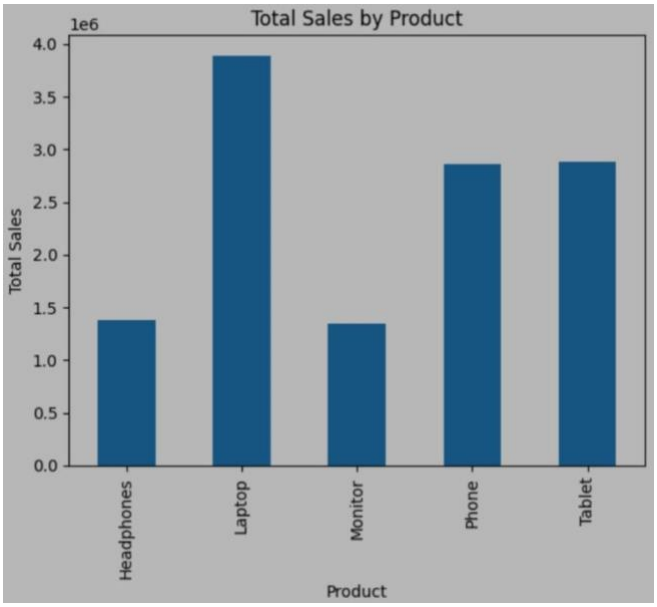
Product_sales = data.groupby(“Product”)[“Total_Sales”].sum()

- Data Visualization

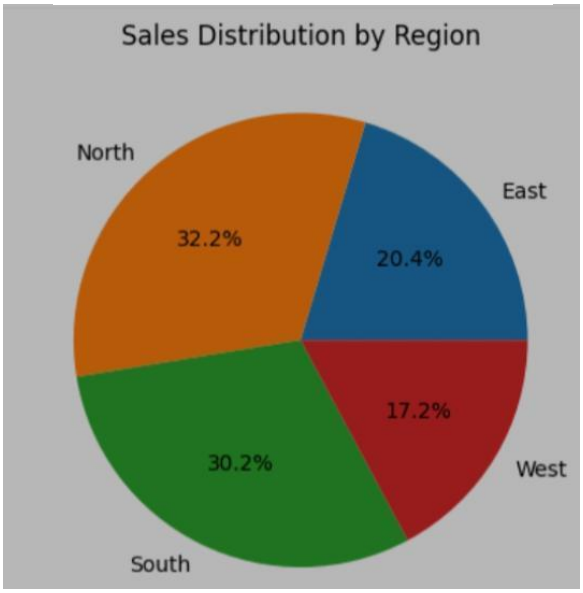
Charts are created using the Matplotlib library to visualize the data and identify trends

VISUAL DOCUMENTATION

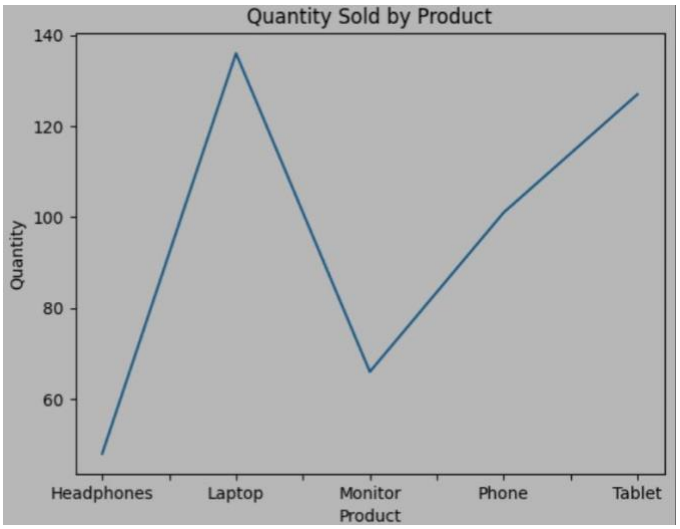
1. Bar chart



2. Pie chart



3. Line chart



Technical Details

Algorithms Used

The project uses simple data analysis techniques such as:

Data grouping

Summation

Aggregation

These techniques help analyze data based on product and region.

Data Structures Used

The following data structures are used in the project:

Pandas DataFrame for storing dataset

Pandas Series for aggregated results

Python lists and dictionaries internally

Architecture

The project follows a simple data analysis workflow:

Dataset → Data Loading → Data Cleaning → Data Analysis → Data Visualization → Insights

Testing Evidence

Several tests were performed to ensure the project works correctly.

Test Case 1 – Dataset Loading

The dataset was successfully loaded using the read_csv function.

Test Case 2 – Missing Value Check

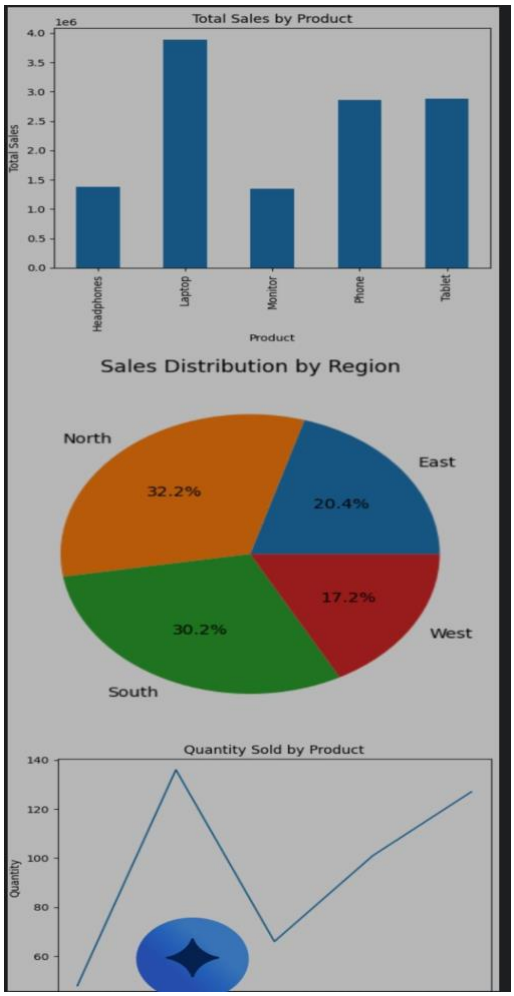
The dataset was checked for missing values using the isnull() function.

Test Case 3 – Chart Generation

Bar charts, pie charts, and line charts were successfully generated using Matplotlib.

All components of the project executed without errors.

```
         Date     Product  Quantity  Price Customer_ID Region  Total_Sales
0  2024-01-01       Phone         7  37300     CUST001   East       261100
1  2024-01-02  Headphones         4  15406     CUST002  North        61624
2  2024-01-03       Phone         2  21746     CUST003   West        43492
3  2024-01-04  Headphones         1  30895     CUST004   East        30895
4  2024-01-05      Laptop         8  39835     CUST005  North       318680
Date           0
Product        0
Quantity       0
Price          0
Customer_ID    0
Region         0
Total_Sales    0
dtype: int64
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 7 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Date         100 non-null    object
 1   Product      100 non-null    object
 2   Quantity     100 non-null    int64
 3   Price        100 non-null    int64
 4   Customer_ID  100 non-null    object
 5   Region       100 non-null    object
 6   Total_Sales  100 non-null    int64
dtypes: int64(3), object(4)
memory usage: 5.6+ KB
None
```



Conclusion

This project demonstrates how Python can be used for data analysis and visualization. Using Pandas and Matplotlib, the sales dataset was explored, analyzed, and visualized effectively.

The project highlights the importance of data visualization in understanding patterns and trends. Such analysis can help businesses improve decision-making and optimize sales strategies.

Overall, this project provides a basic understanding of sales data analysis using Python.